Supplementary Material

A Reproducible MACE Graph-Neural-Network Interatomic Potential for Aspirin: Accuracy, Reliability, and Deployment on Accessible Cloud Infrastructure

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This document provides the complete computational details required to reproduce the results reported in the main text: the software environment, the dataset and data-split protocol, the full model and training configuration, the molecular-dynamics settings, the evaluation-metric definitions, and tabulated numerical results.

S1 Software Environment

All computations used open-source software running on freely available cloud GPU infrastructure (Google Colab; NVIDIA T4 GPU). Package versions are listed in Table S1.

Table S1: Software environment and versions.

Component	Version
Python	3.10
PyTorch	2.1 (CUDA 12.8)
MACE	0.3.16
Atomic Simulation Environment (ASE)	3.28
NumPy / Matplotlib / Seaborn	2.0 / 3.8 / 0.13
Hardware	NVIDIA T4 GPU

S2 Dataset and Data-Split Protocol

Reference energies and atomic forces were taken from the MD17 dataset for aspirin ($C_9H_8O_4$, 21 atoms), comprising 211 762 configurations sampled from 500 K *ab initio* molecular dynamics at the PBE/def2-SVP level. Energies and forces were converted to eV and eV $Å^{-1}$. A single random permutation with fixed seed 42 was used to partition the data; all split index arrays were archived prior to training to guarantee reproducibility and prevent leakage between splits (Table S2).

Table S2: Data-split protocol (seed 42).

Split	Configurations
Training	950
Validation	50
Test (held-out)	210 762
Total	211 762

S3 Model Architecture and Training Configuration

The MACE equivariant graph-neural-network architecture and optimization settings are given in Table S3. These values follow published configurations for organic molecules on MD17.

Table S3: MACE model and training hyperparameters.

Hyperparameter	Value
Message-passing interaction layers	2
Feature channels	128
Maximum angular momentum, L	1
Radial cutoff, r_c	5.0 Å
Correlation order (body order)	3 (4)
Optimizer	AMSGrad
Initial learning rate	0.01
Weight decay	5×10^{-7}
Exponential moving-average decay	0.99
Batch size	10
Force : energy loss weighting	10 : 1
Training epochs	500
Best checkpoint (epoch)	493
Random seed	42

A representative training invocation using the MACE command-line interface is:

```
mace_run_train \
  --name aspirin_mace \
  --train_file aspirin_train.xyz \
  --valid_file aspirin_val.xyz \
  --test_file  aspirin_test.xyz \
  --energy_key energy --forces_key forces \
  --model MACE --num_interactions 2 \
  --hidden_irreps '128x0e+128x1e' \
  --max_ell 1 --r_max 5.0 --correlation 3 \
  --batch_size 10 --max_num_epochs 500 \
  --amsgrad --lr 0.01 --weight_decay 5e-7 \
  --ema --ema_decay 0.99 \
  --energy_weight 1.0 --forces_weight 10.0 \
```

```
--seed 42 --device cuda
```

S4 Molecular-Dynamics Settings

The trained checkpoint was loaded as an ASE calculator. Dynamics used a Langevin thermostat with timestep 0.5 fs and friction coefficient $0.002\,\mathrm{fs}^{-1}$. Energy conservation was verified in the NVE ensemble over 200 steps; a production NVT trajectory was run at 500 K for 250 fs.

S5 Evaluation-Metric Definitions

For M configurations of N atoms with predicted forces $\boldsymbol{F}_{mi}$ and references $\hat{\boldsymbol{F}}_{mi}$:

$$\mathrm{RMSE}_F = \sqrt{\frac{1}{3NM}\sum_m \sum_i \|\boldsymbol{F}_{mi} - \hat{\boldsymbol{F}}_{mi}\|^2}, \tag{1}$$

$$\mathrm{MAE}_F = \frac{1}{3NM}\sum_m \sum_i \sum_\alpha |F_{mi\alpha} - \hat{F}_{mi\alpha}|, \tag{2}$$

with Q_{95} the 95th percentile of the per-component absolute error and r the Pearson correlation between per-configuration force error and reference configuration energy.

S6 Tabulated Numerical Results

Table S4: Training-convergence trajectory at representative epochs.

Epoch	Force RMSE (meV Å^{-1})	Energy RMSE/atom (meV atom^{-1})	Total loss
0	248.10	16.176	0.061579
50	26.44	0.659	0.000699
100	19.83	0.575	0.000393
150	17.48	0.578	0.000306
200	16.30	0.494	0.000266
250	15.72	0.436	0.000247
300	15.20	0.408	0.000231
350	14.86	0.397	0.000221
400	14.62	0.385	0.000214
493	**14.22**	**0.382**	**0.000202**
499	14.29	0.374	0.000204

S7 Data and Code Availability

The training and evaluation scripts, the seed-42 data-split index arrays, the trained model checkpoint, and the complete reproducible pipeline are archived and openly available at [Zenodo DOI to be inserted upon deposit], with the development version maintained at [GitHub repository URL].

Table S5: Full test-set accuracy and generalization metrics.

Metric	Value
Force RMSE (validation)	$14.22\,\mathrm{meV\,Å^{-1}}$
Force RMSE (test)	$15.30\,\mathrm{meV\,Å^{-1}}$
Force MAE (test)	$10.19\,\mathrm{meV\,Å^{-1}}$
Force Q_{95} (test)	$31.26\,\mathrm{meV\,Å^{-1}}$
Relative force RMSE	1.26 %
Energy RMSE/atom (test)	$0.429\,\mathrm{meV\,atom^{-1}}$
Energy MAE/atom (test)	$0.332\,\mathrm{meV\,atom^{-1}}$
Energy Q_{95}/atom (test)	$0.824\,\mathrm{meV\,atom^{-1}}$
Total weighted loss	2.02×10^{-4}
Generalization gap (force)	$1.08\,\mathrm{meV\,Å^{-1}}$ (7.6 %)
Generalization gap (energy)	$0.047\,\mathrm{meV\,atom^{-1}}$ (12.4 %)

Table S6: Force RMSE resolved by spatial component (equivariance test).

Component	x	y	z	Overall
Force RMSE ($\mathrm{meV\,Å^{-1}}$)	16.65	14.98	14.31	15.35

Table S7: Force prediction error by chemical element.

Element	Atom count	Force RMSE ($\mathrm{meV\,Å^{-1}}$)	Force MAE ($\mathrm{meV\,Å^{-1}}$)
Hydrogen (H)	8	8.44	5.43
Carbon (C)	9	18.52	13.37
Oxygen (O)	4	17.94	12.48

Table S8: Computational cost per aspirin configuration and cumulative force-error bounds.

Quantity	Value
Classical force field	0.10 ms
MACE (this work)	0.21 ms
DFT (reference)	5.0 s
Speed-up vs DFT	23 498×
90% of force errors within	$23.5\,\mathrm{meV\,Å^{-1}}$
95% of force errors within	$31.3\,\mathrm{meV\,Å^{-1}}$
99% of force errors within	$52.6\,\mathrm{meV\,Å^{-1}}$

Table S9: Simulated vs reference bond lengths from the NVT trajectory.

Bond	Simulated (Å)	Reference (Å)	Deviation
C−H	1.101	1.09	1.0 %
C−C	1.431	1.40	2.2 %
C−O	1.325	1.36	2.6 %